

# Rebuilding the Deliberation Model from Primitives: Derivations, Proofs, and an Honest Audit\*

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## Abstract

The referee’s central objection is that the paper’s model asserts rather than derives its three headline objects: interior vote margins, posting volume proportional to  $p(1-p)$ , and the welfare kernel  $4p(1-p)(q-\frac{1}{2})^2$ . This note rebuilds the model as a unified two-tier structure — informed contributors post; a continuum of taste-heterogeneous voters decides whether to read and then votes — in which *all* equilibrium objects (readership, posting volume, vote margins, decision quality, and the AI tipping point) are derived from one set of primitives. The main findings: (i) Proposition 1 of the paper ( $\text{Cov}(\text{posts}, \text{margin}) < 0$ ) is now a theorem; (ii) the career-concerns posting motive in the fix roadmap is *provably wrong-signed* for any increasing reputation payoff and must be replaced by an attention/value-of-information motive; (iii) the paper’s reading rule  $V(\pi) = (1-\pi)I_H - c_r$  is derived *exactly*, with  $I_H = 2p(1-p)(2q-1)$ ; (iv) the welfare formula  $4p(1-p)(q-\frac{1}{2})^2$  does *not* survive — the exact value of the forum is  $\min(p, 1-p)$  on an interior “flip window,” zero outside; (v) the tipping-point result survives and is *strengthened*: with endogenous contamination, a positive reading cost alone (no S-shaped adoption curve needed) makes collapse discontinuous, hysteretic, and located at a realized AI share strictly below the reading threshold; (vi) of the calibration, the bound  $\pi^* \geq 0.082$  and the 19.4 words-per-post surplus survive; the  $\times 44.2$  aggregation to 858 words/delegate/proposal survives only under an explicitly stated non-redundancy assumption. A verdict for the Thursday meeting is in Section 12.

## 1 Purpose and the three objections

The automated review makes three related objections to Section 5 of the paper:

- (O1) a single representative delegate making a binary vote cannot generate an interior, continuously varying margin of victory;
- (O2) with a nonatomic continuum of contributors no one is pivotal, so costly posting is not an equilibrium and “posting volume” has no microfoundation;
- (O3) the welfare formula  $W^* - W^{AI} = 4p(1-p)(q-\frac{1}{2})^2$  is asserted, not derived.

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\*Internal theory note for Hall & Huang, “The Effect of AI Writing on Governance: Evidence from a \$3.5 Billion DAO.” Prepared for the Thursday meeting (Decision 1 of the post-review agenda). Builds on V. Huang’s memos `model_derivation_notes.md` and `model_proofs.md`; all results here are re-derived from scratch and verified numerically (script `check_model.py`). Every result is labeled PROVED, PROVED UNDER ASSUMPTIONS, or SKETCH.

All three share one root: the paper’s model has one decision-maker where the data have populations. The rebuild below makes both sides populations. A continuum of privately informed *contributors* decides whether to post; a continuum of *voters* with heterogeneous private tastes decides whether to *read* and then votes. Margins are interior because voters disagree about values, not only facts (answering O1); posting is an equilibrium because posters are paid in *audience*, which is a positive mass even for a measure-zero poster (answering O2, with the assumption doing the work stated explicitly); and welfare is computed exactly from the equilibrium voting behavior (answering O3 — with the consequence that the old formula must be replaced).

Two candidate posting motives were on the table in the fix roadmap. We prove below (Theorem 1) that the reputation/career-concerns motive generates posting volume *minimized* on contested proposals — the opposite of the data — for *any* increasing reputation payoff, and therefore cannot underlie Proposition 1. The attention/value-of-information motive works and is adopted.

## 2 The model

### 2.1 Primitives

A DAO evaluates one proposal. Its unknown quality is  $\theta \in \{G, B\}$  with common prior  $p \equiv \Pr(\theta = G) \in (0, 1)$ . Write  $x \equiv p(1 - p)$  and  $d \equiv |p - \frac{1}{2}|$ ;  $d$  measures ex-ante *consensus* ( $d$  large) versus *contestedness* ( $d$  small).

**Contributors.** A continuum of mass 1 of potential posters. Contributor  $i$  observes a private signal  $s_i \in \{g, b\}$  with symmetric precision  $q \equiv \Pr(s_i = g | G) = \Pr(s_i = b | B) \in (\frac{1}{2}, 1)$ , i.i.d. conditional on  $\theta$ , and draws a posting cost  $c_i \sim F$  on  $[0, \bar{c}]$ , independent of  $s_i$ , where  $F$  is continuous and strictly increasing with  $F(0) = 0$ . Posting publishes a message that reveals  $s_i$ .

**AI content.** A mass  $z \geq 0$  of AI-generated posts is present in the thread; an AI post’s message is uninformative ( $\Pr(\text{message} = g | \theta) = \frac{1}{2}$  for both  $\theta$ ) and is observationally indistinguishable from a human post. If human posting mass is  $n$ , the fraction of AI posts is  $\pi = z/(z + n)$ . Sections 3–7 treat  $\pi$  as a parameter; Section 9 makes it an equilibrium object.

**Voters.** A continuum of mass 1. Voter  $j$  has a private taste  $b_j \sim \text{Unif}[-a, a]$ , i.i.d., orthogonal to  $\theta$  (idiosyncratic value from the proposal passing: exposure, vested interests, ideology). Common values are  $v(G) = +1$ ,  $v(B) = -1$ : voting *For* a proposal that passes yields  $v(\theta) + b_j$ , *Against* yields 0. Before voting, a voter may pay  $c_r \geq 0$  to read one post sampled uniformly at random from the thread (Remark 6 discusses multi-post reading).

**Timing.** (1) Nature draws  $\theta$ , signals, costs, tastes. (2) Contributors post or not; AI mass  $z$  is posted. (3) Each voter chooses to read or not; readers sample one post each, independently. (4) All voters vote For or Against; the proposal passes iff the For-share  $\Phi$  exceeds  $\frac{1}{2}$ . (5) Payoffs realize.

### 2.2 Assumptions, stated honestly

**Assumption 1** (Exact LLN). *Conditional on  $\theta$ , cross-sectional averages of i.i.d. individual quantities equal their conditional means (the usual continuum convention; see Judd 1985 for the measurability caveat). All “deterministic” aggregates below are exact in this convention and hold up to  $O(N^{-1/2})$  noise with  $N$  finite agents.*

**Assumption 2** (Decision-theoretic voting and reading). *Each voter votes — and values pre-vote information — as if her vote determined the outcome for her own payoff: she votes For iff  $\mathbb{E}[v(\theta) \mid \text{her information}] + b_j \geq 0$ , and values reading by the improvement in this decision.*

This is the standard sincere/expressive-voting device for large elections (e.g., ethical-voter models in the spirit of Feddersen and Sandroni 2006). It must be said plainly: *the pivotality problem on the voter side is bypassed by this assumption, not solved*. What the assumption buys is discipline: given it, everything else (who reads, how vote shares move, when the forum dies) is derived. The alternative — fully instrumental voters — makes reading and voting trivially worthless for every atomless agent and cannot generate any of the observed cross-sectional structure.

**Assumption 3** (Truthful posting). *A post reveals the contributor’s signal truthfully.*

Justifiable by reputational cheap-talk arguments (posts are public and  $\theta$  is eventually revealed; systematic misreporting is detectable ex post, cf. Ottaviani and Sørensen 2006), but in this note it is an assumption.

**Assumption 4** (Attention rewards). *A contributor who posts receives  $R \cdot A$ , where  $A$  is the mass of readers of the thread and  $R > 0$  converts audience into utility.*

This is the posting motive that survives the continuum: audience is a positive *mass* even though any single poster is measure zero, so no pivotality is needed. Microfoundations consistent with the setting: delegate incentive programs that compensate visible participation; attracting delegated voting power from readers; direct attention utility. What Assumption 4 deliberately does *not* assume is that posting signals ability — Theorem 1 shows that motive predicts the wrong sign. Remark 1 shows the results are robust to rewarding *per-post* rather than *per-thread* attention.

**Assumption 5** (Interior tastes).  *$a \geq 1$ , so every posterior-based voting threshold lies inside  $[-a, a]$  and vote shares are interior.*

**Assumption 6** (Pre-AI viability).  *$c_r < I_H(p) \equiv 2p(1-p)(2q-1)$  for the proposals under study: absent AI, reading has positive net value for at least the most contested proposals.*

### 2.3 Notation and two identities

A sampled post is human with probability  $1 - \pi$ ; hence its message has effective precision

$$Q \equiv \Pr(\text{message} = g \mid G) = (1 - \pi)q + \pi \cdot \frac{1}{2}, \quad \text{so} \quad Q - \frac{1}{2} = (1 - \pi)\left(q - \frac{1}{2}\right). \quad (1)$$

This is the “signal dilution” equation of the working notes, here *derived* from the sampling protocol rather than posited. Write

$$D_1 = \Pr(g) = pQ + (1-p)(1-Q), \quad D_2 = \Pr(b) = p(1-Q) + (1-p)Q, \quad \mu_G = \frac{pQ}{D_1}, \quad \mu_B = \frac{p(1-Q)}{D_2},$$

for the message probabilities and posteriors ( $\mu_m = \Pr(G \mid m)$ ), and  $D \equiv D_1 D_2$ .

**Lemma 1** (Bayesian identities [PROVED]).

$$(I1) \quad D_1(\mu_G - p) = D_2(p - \mu_B) = p(1-p)(2Q-1), \quad (2)$$

$$(I2) \quad \mu_G - \mu_B = \frac{p(1-p)(2Q-1)}{D_1 D_2}, \quad (3)$$

$$(I3) \quad D_1 D_2 = Q(1-Q) + p(1-p)(2Q-1)^2. \quad (4)$$

*Proof.* (I1):  $D_1\mu_G = pQ$ , so  $D_1(\mu_G - p) = pQ - pD_1 = p[(1-p)Q - (1-p)(1-Q)] = x(2Q-1)$ ; symmetrically for  $D_2$ . (I2):  $\mu_G - \mu_B = (\mu_G - p) + (p - \mu_B) = x(2Q-1)(D_1^{-1} + D_2^{-1})$  and  $D_1 + D_2 = 1$ . (I3): expand  $D_1D_2 = Q(1-Q)[p^2 + (1-p)^2] + x[Q^2 + (1-Q)^2] = Q(1-Q)(1-2x) + x[1-2Q(1-Q)]$  and collect terms.  $\square$

Note  $D \in (0, \frac{1}{4}]$ , with  $D = \frac{1}{4}$  at  $p = \frac{1}{2}$ .

### 3 The reading stage: who reads, and the paper's reading rule derived

Voters differ in how useful the forum is to them. A voter who does not read votes on the prior: For iff  $b_j \geq t_0 \equiv 1 - 2p$ . A reader who sees message  $m$  votes For iff  $b_j \geq t_m \equiv 1 - 2\mu_m$ ; note  $t_G < t_0 < t_B$ .

**Lemma 2** (Value of reading is a tent [PROVED]). *The value of information from reading one post for a voter with taste  $b$  is*

$$\text{VOI}(b) = \begin{cases} D_1(b - t_G), & b \in [t_G, t_0], \\ D_2(t_B - b), & b \in [t_0, t_B], \\ 0, & \text{otherwise:} \end{cases}$$

*a tent on the swing band  $[t_G, t_B]$  (base  $2(\mu_G - \mu_B)$ ), with peak at the ex-ante indifferent voter  $b = t_0$  of height*

$$h(p, Q) = 2p(1-p)(2Q-1). \quad (5)$$

*Proof.* Reading changes the action only for  $b \in [t_G, t_B]$ . For  $b \in [t_G, t_0]$  the voter switches to For only on  $m = g$ , gaining  $D_1[(2\mu_G - 1) + b] = D_1(b - t_G) \geq 0$ ; for  $b \in [t_0, t_B]$  she switches to Against only on  $m = b$ , gaining  $D_2[-(2\mu_B - 1) - b] = D_2(t_B - b) \geq 0$ . Both pieces vanish at the band edges and meet at  $b = t_0$ , where by (I1) each equals  $2D_1(\mu_G - p) = 2x(2Q-1) = h$ .  $\square$

**Proposition 1** (Readership [PROVED]). *A voter reads iff  $\text{VOI}(b_j) \geq c_r$ . The reading set is the band  $[\ell, u]$  with  $\ell = t_G + c_r/D_1$ ,  $u = t_B - c_r/D_2$  (empty iff  $h \leq c_r$ ), and its mass is*

$$A(p; Q, c_r) = \frac{(2p(1-p)(2Q-1) - c_r)_+}{2a D_1 D_2} = \frac{\mu_G - \mu_B}{a} \left(1 - \frac{c_r}{h}\right)_+. \quad (6)$$

*A is symmetric in  $p$  about  $\frac{1}{2}$  and strictly increasing in  $x = p(1-p)$  wherever positive; hence single-peaked at  $p = \frac{1}{2}$  and zero for consensus proposals. Moreover, for the peak (ex-ante indifferent) voter, the net value of reading is exactly*

$$V(\pi) = (1 - \pi) I_H - c_r, \quad I_H \equiv h(p, q) = 2p(1-p)(2q-1), \quad (7)$$

*so she reads iff  $\pi \leq \bar{\pi}(p) \equiv 1 - c_r/I_H(p)$ .*

*Proof.* The set  $\{\text{VOI} \geq c_r\}$  is the tent's cross-section at height  $c_r$ ; for a triangle of peak  $h$ , each side's width scales by  $(1 - c_r/h)$ , giving mass  $(t_B - t_G)(1 - c_r/h)_+/(2a)$ ; apply (I2), and (I1)+(I3) for the first form. Monotonicity in  $x$ : differentiating (6) with  $D = Q(1-Q) + x(2Q-1)^2$ ,

$$\frac{\partial A}{\partial x} = \frac{2(2Q-1)Q(1-Q) + c_r(2Q-1)^2}{2a D^2} > 0.$$

For (7): by Lemma 2 the peak voter's gross value is  $h(p, Q) = 2x(2Q - 1)$ , and by (1),  $2Q - 1 = (1 - \pi)(2q - 1)$ , so  $h(p, Q) = (1 - \pi)h(p, q) = (1 - \pi)I_H$ .  $\square$

Equation (7) is the paper's equation  $V(\pi) = (1 - \pi)I_H - c_r$  and its threshold  $\pi^* = 1 - c_r/I_H$  — previously asserted for a representative delegate, now derived exactly for the marginal (peak) reader, with the previously free parameter  $I_H$  pinned to primitives:  $I_H = 2p(1 - p)(2q - 1)$ . Three consequences:

- *The threshold is proposal-specific:*  $\bar{\pi}(p) = 1 - c_r/[2p(1 - p)(2q - 1)]$ . Consensus proposals (low  $x$ ) have low thresholds and stop being read first; knife-edge proposals are read the longest. “The forum dies from the tails inward.”
- *Non-peak readers* have strictly smaller gross value (the tent), so each voter has her own threshold below  $\bar{\pi}(p)$ ; aggregate readership (6) declines in  $\pi$  and reaches *exactly zero* at the interior contamination level  $\bar{\pi}(p) < 1$ .
- *Readership falls with contamination:* with  $c_r = 0$ ,  $A = x(2q - 1)(1 - \pi)/(aD)$  — near-linear in  $(1 - \pi)$  (exact numerator linearity; the denominator's  $\pi$ -dependence is second order). This is the working notes' new testable prediction; it survives the rebuild.

## 4 The posting stage: volume, and why career concerns fail

**Proposition 2** (Posting volume [PROVED UNDER ASSUMPTIONS] (Assumptions 3, 4)). *Contributor  $i$  posts iff  $c_i \leq RA(p)$ , so equilibrium human posting mass is*

$$n(p) = F(RA(p)), \quad (8)$$

which inherits the shape of  $A$ : symmetric about  $p = \frac{1}{2}$ , single-peaked there, zero on consensus proposals with  $h \leq c_r$ , and nonincreasing in  $d = |p - \frac{1}{2}|$  (strictly decreasing wherever  $0 < RA < \bar{c}$ ). If  $F$  is near-linear at the origin and  $c_r$  is small,

$$n(p) \propto A(p) \approx \frac{2p(1 - p)(2Q - 1)}{a} \cdot \frac{1}{4D} \propto p(1 - p),$$

up to the mild factor  $\frac{1}{4}D^{-1} \in [1, (4Q(1 - Q))^{-1}]$ .

*Proof.* Immediate from Assumption 4 and Proposition 1. The approximation uses  $D \rightarrow \frac{1}{4}$  as  $p \rightarrow \frac{1}{2}$  or  $Q \rightarrow \frac{1}{2}$ .  $\square$

So “posting volume  $\propto p(1 - p)$ ” is delivered as a first-order approximation to an exact closed form, with the exact form available for structural work. Posting peaks on contested proposals *because that is where the audience is*, and the audience is there because that is where information has value. One force (the tent peaking at  $p = \frac{1}{2}$ ) drives reading, posting, and — Section 7 — welfare.

*Remark 1* (Congestion variant). If instead each of the  $n$  posts receives only its *share* of attention (reward  $RA/n$ , readers spread over posts), the equilibrium condition becomes  $n = F(RA/n)$ ; with  $F$  linear at the origin,  $n \propto \sqrt{A}$ . All shape results below need only that  $n$  is a strictly increasing transform of  $A$ , so Theorem 3 is unaffected; only the exact locus ( $p(1 - p)$  vs. its square root) changes — a distinction the data cannot make anyway (Section 11).

#### 4.1 The negative result: reputation for ability posts on the wrong proposals

The fix roadmap proposed career concerns — posting to attract delegated voting power by signaling ability — as the posting motive. We formalize it and prove it fails, robustly.

Suppose contributors have ability  $\alpha \in \{H, L\}$ ,  $\Pr(H) = \gamma$ , with signal precisions  $q_H > q_L > \frac{1}{2}$ ; a contributor does not observe her own type (the standard career-concerns case, Holmström 1999). Delegators observe a poster's message and, ex post,  $\theta$ , and update; non-posters retain reputation  $\gamma$  (no inference from silence; see the remark below). The poster's payoff is  $\phi(\hat{\gamma}) - c_i$  where  $\hat{\gamma}$  is her posterior reputation and  $\phi$  is *any* strictly increasing function.

**Theorem 1** (Career concerns are wrong-signed [PROVED]). *Let  $\bar{q} = \gamma q_H + (1 - \gamma)q_L$ . Then:*

(a) *Posting leads to exactly two possible reputations, independent of  $p$ :*

$$\begin{aligned} r_+ &= \frac{\gamma q_H}{\gamma q_H + (1 - \gamma)q_L} > \gamma && \text{(vindicated, } s = \theta), \\ r_- &= \frac{\gamma(1 - q_H)}{\gamma(1 - q_H) + (1 - \gamma)(1 - q_L)} < \gamma && \text{(wrong, } s \neq \theta). \end{aligned}$$

(b) *The expected gain from posting signal  $s$  is*

$$g(s, p) = \rho(s, p) \phi(r_+) + [1 - \rho(s, p)] \phi(r_-) - \phi(\gamma),$$

*strictly increasing in the vindication probability  $\rho(s, p) = \Pr(\theta = s \mid s)$ .*

(c) *At  $p = \frac{1}{2}$ ,  $\rho(s, \frac{1}{2}) = \bar{q}$  for both signals; as  $p \rightarrow 1$ ,  $\rho(g, p) \rightarrow 1$ . Hence for every strictly increasing  $\phi$ , the per-contributor posting gain at a foregone conclusion (majority signal) strictly exceeds the gain at a toss-up, and total posting mass satisfies*

$$n^{\text{cc}}(\frac{1}{2}) < \lim_{p \rightarrow 1} n^{\text{cc}}(p).$$

*For linear  $\phi$  the gain at  $p = \frac{1}{2}$  is exactly zero (so  $n^{\text{cc}}(\frac{1}{2}) = 0$ ), and numerically  $n^{\text{cc}}(p)$  is V-shaped in  $p$  throughout the parameter range checked.*

*Consequently career-concerns posting concentrates on consensus proposals, generating a nonnegative posts-margin covariance — the opposite of the data — and cannot rationalize the paper's Proposition 1.*

*Proof.* (a) Given  $(s, \theta)$  the type-likelihood is  $\Pr(s \mid \theta, \alpha)$ , which by symmetry of the precisions equals  $q_\alpha$  if  $s = \theta$  and  $1 - q_\alpha$  otherwise — free of  $p$ . Bayes gives  $r_\pm$ ;  $r_+ > \gamma > r_-$  follows from  $q_H > q_L$ . (b) Immediate from (a). (c)  $\rho(s, p) = \Pr(\theta = s \mid s)$  with the mixture precision  $\bar{q}$ : at  $p = \frac{1}{2}$ ,  $\rho = \bar{q}$  for both signals; as  $p \rightarrow 1$ ,  $\rho(g, p) = p\bar{q}/[p\bar{q} + (1 - p)(1 - \bar{q})] \rightarrow 1$  and the population share of  $g$ -signals  $\rightarrow 1$ . Since  $\rho(g, \cdot)$  rises from  $\bar{q}$  to 1 and  $\phi$  is strictly increasing,  $g(g, p) \rightarrow \phi(r_+) - \phi(\gamma) > g(s, \frac{1}{2})$  for both  $s$ ; posting mass is  $\sum_s \Pr(s)F(g(s, p)_+)$ , giving the strict ranking. For linear  $\phi$ : by iterated expectations  $\mathbb{E}_\theta[\Pr(H \mid s, \theta) \mid s] = \Pr(H \mid s)$ , and at  $p = \frac{1}{2}$  the signal is uninformative about type ( $\Pr(s \mid \alpha) = \frac{1}{2}$  for both  $\alpha$ ), so  $\Pr(H \mid s) = \gamma$  and  $g = 0$ .  $\square$

The economics: on a toss-up your signal is (nearly) a coin flip *whatever your ability*, so being right is barely diagnostic in expectation — and the reputational payoffs  $r_\pm$  from vindication are the same everywhere, while the *probability* of vindication is maximal when you hold the majority

view on a foregone conclusion. Career concerns therefore push posting toward safe consensus, not contested debate. This kills the roadmap's step (ii) as stated, and it also means the paper's delegate posting  $\leftrightarrow$  voting-power correlation should *not* be narrated as career-concerns evidence.

*Remark 2* (Robustness and caveats of the negative result). The proof covers arbitrary increasing  $\phi$  (so convex, tournament-style delegation markets do not rescue the mechanism; numerically the V-shape persists for  $\phi(r) = r^3$  and  $\phi = \sqrt{r}$ , with the minimum near — though for convex  $\phi$  not exactly at —  $p = \frac{1}{2}$ ). Two modeling choices are substantive: non-posters are not updated on (equilibrium inference from silence is not modeled), and contributors do not know their own type. Relaxing either changes the game but not the core force, which is (a): vindication payoffs are  $p$ -invariant while vindication probability rises with consensus. We label the endpoint comparison PROVED and the global V-shape NUMERICALLY VERIFIED.

## 5 The voting stage: interior margins, derived

Now aggregate votes. Conditional on  $\theta$ , each reader's sampled message is i.i.d. ( $g$  with probability  $\rho_G = Q$  if  $\theta = G$ ;  $\rho_B = 1 - Q$  if  $\theta = B$ ), so by Assumption 1 the For-share is deterministic given  $\theta$ :

$$\Phi(\theta) = \underbrace{\frac{a - u}{2a}}_{\text{non-readers with } b > u} + \underbrace{A \cdot \rho_\theta}_{\text{readers voting For}}, \quad (\text{non-readers with } b < \ell \text{ vote Against}),$$

because within the reading band a  $g$ -message moves every reader to For and a  $b$ -message to Against. The *realized margin* is  $M(\theta) = |2\Phi(\theta) - 1|$  and the expected margin is  $\mathbb{E}[M \mid p] = p M(G) + (1 - p) M(B)$ .

**Proposition 3** (Interior margins, closed form [PROVED]). *Let  $m_\theta \equiv 2\Phi(\theta) - 1$ . In the active region ( $h > c_r$ ),*

$$a D m_G = Q(1 - Q)(2p - 1) + (2Q - 1)^2 x - c_r(1 - p)(2Q - 1), \quad (9)$$

$$a D m_B = Q(1 - Q)(2p - 1) - (2Q - 1)^2 x + c_r p(2Q - 1), \quad (10)$$

and when  $h \leq c_r$  (no readers)  $m_G = m_B = (2p - 1)/a$ . For  $p \geq \frac{1}{2}$ ,  $m_G \geq 0$  always; margins are interior (in  $(0, 1)$ ) for all interior  $\mu$  by Assumption 5. Under  $p \mapsto 1 - p$  the model is symmetric:  $m_B(1 - p) = -m_G(p)$ .

*Proof.* With band  $[\ell, u]$ ,  $\ell = t_G + c_r/D_1$ ,  $u = t_B - c_r/D_2$ : every reader votes For on message  $g$  and Against on message  $b$  (the band lies inside  $[t_G, t_B]$ ), so  $m_\theta = -u/a + 2A\rho_\theta$ . Substitute  $u = 1 - 2\mu_B - c_r/D_2$  and (6), write  $2\mu_B - 1 = (\mu_G + \mu_B - 1) - (\mu_G - \mu_B)$ , and use (I2) together with three one-line identities:  $\mu_G + \mu_B - 1 = \mu_G - (1 - \mu_B) = Q[pD_2 - (1 - p)D_1]/D$  with  $pD_2 - (1 - p)D_1 = (1 - Q)[p^2 - (1 - p)^2] = (1 - Q)(2p - 1)$ , so  $\mu_G + \mu_B - 1 = Q(1 - Q)(2p - 1)/D$ ; and  $D_1 - Q = -(1 - p)(2Q - 1)$ ,  $D_1 - (1 - Q) = p(2Q - 1)$  for the  $c_r$  terms. Sign of  $m_G$  for  $p \geq \frac{1}{2}$ : the negative term obeys  $c_r(1 - p)(2Q - 1) < 2x(2Q - 1)^2(1 - p) \leq (2Q - 1)^2 x$  using  $c_r < h$  and  $2(1 - p) \leq 1$ . Symmetry: substitute  $p \rightarrow 1 - p$  in (10) ( $x, D$  invariant). All formulas verified against direct numerical integration of the taste distribution (300 random parameter draws, max abs. error  $< 4 \times 10^{-7}$ ).  $\square$

**Theorem 2** (Expected margin rises with consensus [PROVED]).  $\mathbb{E}[M \mid p]$  is continuous, symmetric about  $p = \frac{1}{2}$ , and strictly increasing in  $d = |p - \frac{1}{2}|$  on  $(0, \frac{1}{2})$ , for all  $Q \in (\frac{1}{2}, 1]$ ,  $a \geq 1$ ,  $c_r \geq 0$ . Moreover

$$\mathbb{E}[M \mid p] \geq \frac{|2p - 1|}{a}, \quad (11)$$

with equality outside an interior contested window (in particular everywhere once readership is zero), and  $\mathbb{E}[M \mid \frac{1}{2}] = (2Q - 1)^2(1 - \frac{2c_r}{2Q-1})/a \geq 0$  at the center (with  $c_r = 0$ :  $(2Q - 1)^2/a$ ).

*Proof.* Take  $p \geq \frac{1}{2}$  (symmetry). Write  $\alpha = Q(1 - Q)(2p - 1)$ ,  $\beta = (2Q - 1)^2x$ ,  $\gamma_G = c_r(1 - p)(2Q - 1)$ ,  $\gamma_B = c_rp(2Q - 1)$ , so  $aDm_G = \alpha + \beta - \gamma_G$ ,  $aDm_B = \alpha - \beta + \gamma_B$ . Three regimes as  $d$  rises from 0:

(iii) *Contested window* ( $m_B < 0$ ): here  $aD\mathbb{E}[M \mid p] = p(\alpha + \beta - \gamma_G) + (1 - p)(\beta - \alpha - \gamma_B) = \alpha(2p - 1) + \beta - [p\gamma_G + (1 - p)\gamma_B]$ , and  $p\gamma_G + (1 - p)\gamma_B = 2c_r(2Q - 1)x$ , so with  $\kappa \equiv (2Q - 1)^2/[Q(1 - Q)]$  and  $\kappa_c \equiv (2Q - 1)[(2Q - 1) - 2c_r]/[Q(1 - Q)] \in (0, \kappa]$  (positive because  $2c_r < 4x(2Q - 1) \leq 2Q - 1$  in the active region),

$$a\mathbb{E}[M \mid p] = \frac{4d^2 + \kappa_c x}{1 + \kappa x}, \quad x = \frac{1}{4} - d^2.$$

Differentiating in  $d^2$ : the numerator of the derivative is  $(4 - \kappa_c)(1 + \kappa x) + \kappa(4d^2 + \kappa_c x) = 4 + \kappa - \kappa_c > 0$  (the  $d^2$ -terms cancel identically). Strictly increasing.

(ii) *Consensus, active* ( $m_B \geq 0$ ,  $h > c_r$ ): both margins positive, and  $aD[p m_G + (1 - p)m_B] = \alpha + \beta(2p - 1) + c_r(2Q - 1)[-(1 - p)p + (1 - p)p] = (2p - 1)D$ , so  $\mathbb{E}[M \mid p] = (2p - 1)/a$  exactly — strictly increasing.

(i) *No readers* ( $h \leq c_r$ ):  $\mathbb{E}[M \mid p] = (2p - 1)/a$ , strictly increasing.

Continuity: at the (iii)/(ii) boundary  $m_B = 0$  both expressions coincide; at the (ii)/(i) boundary  $\ell = u = t_0$  and  $m_\theta \rightarrow (2p - 1)/a$  smoothly. A continuous function that is strictly increasing on each piece of a partition of  $[0, \frac{1}{2})$  into intervals is strictly increasing. The floor (11): in regimes (i)–(ii) it binds; in (iii),  $aD\mathbb{E}[M \mid p] - (2p - 1)D = (1 - 2d)(\beta - \alpha) - 2c_r(2Q - 1)x > (1 - 2d)\gamma_B - 2c_r(2Q - 1)x = c_r(2Q - 1)[2p(1 - p) - 2x] = 0$ , using  $m_B < 0 \iff \beta - \alpha > \gamma_B$ . Monotonicity was additionally verified numerically on a  $7 \times 3 \times 5$  parameter grid with  $2 \times 10^4$  points of  $p$  each: zero violations.  $\square$

Objection (O1) is answered: margins are interior and continuously distributed because voters disagree on *values* ( $a > 0$ ); they are informative about  $p$  because the expected margin strictly increases in consensus  $d$ ; and the paper’s shorthand “margin  $\propto |2p - 1|$ ” holds *exactly* outside the contested window (and as the leading term for weak signals, since  $\mathbb{E}[M \mid \frac{1}{2}] = O((2Q - 1)^2)$ ).

## 6 The paper’s Proposition 1, now a theorem

**Theorem 3** (Posts–margin covariance [PROVED]). *Let proposals draw priors  $p_i$  i.i.d. from any distribution such that  $d_i = |p_i - \frac{1}{2}|$  is nondegenerate and readership  $A(p_i)$  is nonconstant on its support; let  $\theta_i$  be drawn given  $p_i$ , and let  $(\text{Posts}_i, \text{Margin}_i) = (n(p_i), M_i)$  be the realized volume and margin. Then*

$$\text{Cov}(\text{Posts}_i, \text{Margin}_i) < 0.$$

*Proof.*  $n(p_i)$  is  $\sigma(p_i)$ -measurable, so by the law of total covariance  $\text{Cov}(n, M) = \text{Cov}(n(p), \mathbb{E}[M \mid p])$  ( $M$ ’s within- $p$  variation over  $\theta$  contributes nothing). By Propositions 1 and 2,  $n$  is a non-increasing function  $\nu(d)$ ; by Theorem 2,  $\mathbb{E}[M \mid p]$  is a strictly increasing function  $\rho(d)$ . For



i.i.d. copies  $d, d'$ ,  $\text{Cov}(\nu(d), \rho(d)) = -\frac{1}{2}\mathbb{E}[(\nu(d) - \nu(d'))(\rho(d) - \rho(d'))]$ , where every term in the expectation is  $\leq 0$  (opposite monotonicities) and is strictly negative with positive probability because  $\nu$  is nonconstant and  $\rho$  is strictly increasing on a nondegenerate  $d$ .  $\square$

This is objection (O2) resolved on the paper's own terms: the negative covariance between posting volume and margin of victory — the paper's central empirical object — is a consequence of the equilibrium, with the posting equilibrium sustained without any pivotality (Assumption 4) and the margin process derived (Theorem 2). Both conditional moments are monotone transforms of the single latent  $d_i$ , which is what licenses the paper's practice of reading margins as inverse measures of contestedness.

*Remark 3* (The honest ceiling on empirical content). Theorem 3 pins a *sign*, not a locus. On the pre-break sample ( $n = 29$ ) the structural locus and a linear reduced form are statistically indistinguishable (the working notes'  $R^2$ : 0.114 vs 0.111). The model's discriminating content is cross-equation (the same  $q$  appears in (6), (9)–(10), and the welfare window below), and the working notes report those over-identifying tests as underpowered on this sample. Deriving the model buys internal rigor, not additional empirical verification — this should be said in the paper.

## 7 Welfare: the exact value of the forum, and the fate of $4p(1 - p)(q - \frac{1}{2})^2$

The DAO's decision payoff is the probability the outcome matches  $\theta$  (pass iff  $G$ ). Without a forum, the vote aggregates only tastes and the prior: the outcome follows the prior action, correct with probability  $\max(p, 1 - p)$ .

**Proposition 4** (Exact decision value of the forum [PROVED]). *Define the flip window*

$$\Omega(Q, c_r) = \left\{ p : (2Q - 1)^2 p(1 - p) - c_r(2Q - 1) \max(p, 1 - p) > Q(1 - Q) |2p - 1| \right\},$$

a symmetric interval around  $p = \frac{1}{2}$ , nonempty iff  $c_r < \frac{2Q-1}{2}$ , expanding to  $(0, 1)$  as  $Q \rightarrow 1$ ,  $c_r \rightarrow 0$  and shrinking to  $\emptyset$  as  $Q \rightarrow \frac{1}{2}$ . In equilibrium:

- (a) For  $p \in \Omega$ :  $m_G > 0 > m_B$  — the vote passes good proposals and rejects bad ones with probability one (full information aggregation, the Feddersen–Pesendorfer property delivered by the LLN across readers).
- (b) For  $p \notin \Omega$ : the outcome coincides with the prior action.
- (c) Hence the forum's gross decision value is

$$V(p; Q, c_r) = \min(p, 1 - p) \cdot \mathbf{1}\{p \in \Omega(Q, c_r)\}, \quad (12)$$

and net welfare subtracts participation costs  $W = \Pr(\text{correct}) - c_r A(p) - \int_0^{RA(p)} c dF(c)$ .

*Proof.* For  $p \geq \frac{1}{2}$  in the active region,  $m_G > 0$  always (Proposition 3), so correctness turns on  $m_B < 0$ , i.e.  $\beta - \gamma_B > \alpha$ , which is the displayed window condition with  $\max(p, 1 - p) = p$ ; the  $p \leq \frac{1}{2}$  branch is the mirror condition  $m_G > 0$ . Window  $\subset$  active region:  $m_B < 0$  requires  $\beta > \gamma_B$ , i.e.  $x(2Q - 1) > c_r p$ , which for  $p \geq \frac{1}{2}$  implies  $h = 2x(2Q - 1) > 2c_r p \geq c_r$ . Nonemptiness iff the condition holds at  $p = \frac{1}{2}$ :  $(2Q - 1)^2/4 - c_r(2Q - 1)/2 > 0 \iff c_r < (2Q - 1)/2 = h(\frac{1}{2}, Q)$ ,

which is also exactly the condition that *any* proposal has readers. (b): outside  $\Omega$  but active,  $m_G$  and  $m_B$  share the prior’s sign, so the outcome is the prior action; inactive is immediate. (c): conditional correctness is  $p \cdot 1 + (1 - p) \cdot 1 = 1$  inside and  $\max(p, 1 - p)$  outside; subtract the prior baseline.  $\square$

**Verdict on the paper’s formula.** The kernel  $4p(1 - p)(q - \frac{1}{2})^2$  is *not* the value of the forum in any version of the model:

- In the derived two-tier model the exact object is (12): a tent  $\min(p, 1 - p)$  truncated to an interior window — *closeness enters as the level, informativeness as the width*. AI dilution ( $Q \rightarrow \frac{1}{2}$  via (1)) destroys value by *narrowing the window* until it vanishes at  $\bar{\pi}(\frac{1}{2})$ ; the collapse destroys  $\min(p, 1 - p)$  of decision quality per proposal, maximal ( $= \frac{1}{2}$ ) on contested proposals. The qualitative claim the paper builds on — AI destroys the most welfare where governance is most contested — survives exactly.
- For a *single* common signal of precision  $q$  (closest to the old representative-delegate framing), the exact value is the Blackwell tent  $(q - \max(p, 1 - p))_+$ , and the exact belief-accuracy (variance-reduction) value is  $[p(1 - p)(2q - 1)]^2 / (D_1 D_2)$ , which is  $16[p(1 - p)]^2 (q - \frac{1}{2})^2$  to leading order — note the *square* on  $p(1 - p)$ . The paper’s monomial matches neither; it is a smooth mnemonic with the right single-peakedness and the right  $(q - \frac{1}{2})^2$  scaling but no decision-theoretic pedigree.

*Recommendation:* replace the formula with (12) (decision-quality story, matches the rest of the model), or — if a smooth kernel is wanted for prose — present  $4p(1 - p)(q - \frac{1}{2})^2$  explicitly as a heuristic interpolation, never as a result. Since the words-based calibration never used this formula (it runs off  $I_H$ ,  $c_r$ ,  $\pi^*$  — see Section 10), replacing it costs nothing quantitative.

*Remark 4* (Why full aggregation inside the window is the right idealization). Statement (a) is the continuum idealization of a Condorcet effect: finitely many readers make the probability of a correct outcome  $1 - O(e^{-cN})$  inside the window. The knife-edge flavor of “probability one” is Assumption 1, not deep structure; what is robust is that decision quality is high inside the window and prior-driven outside.

## 8 The AI shock: dilution, margin compression, and the reading collapse

AI enters through two channels, both now derived: the sampled-post precision falls (equation (1)), and the reading calculus (7) deteriorates.

**Proposition 5** (Comparative statics in contamination [PROVED]). *As  $\pi$  rises (equivalently  $Q \downarrow \frac{1}{2}$ ):*

- Readership  $A(p; \pi)$  falls, with exactly linear numerator  $(2x(2q - 1)(1 - \pi) - c_r)_+$ , reaching zero at  $\bar{\pi}(p) = 1 - c_r / I_H(p) < 1$ : an interior contamination level at which the forum’s audience is exactly zero. Consensus proposals hit their thresholds first.*
- Posting  $n(p; \pi) = F(RA)$  falls with readership — the audience channel: AI does not need to out-argue humans, it only needs to thin the readership that pays for human effort.*
- Margins compress toward the no-forum benchmark:  $\mathbb{E}[M \mid p]$  is nondecreasing in  $Q$  pointwise, falling to the floor  $|2p - 1|/a$ ; the contested window  $\Omega$  narrows to  $\emptyset$ .*

- (d) *Welfare*:  $V(p; Q_{\text{eff}}, c_r)$  collapses window-first; at  $\pi \geq \bar{\pi}(\frac{1}{2})$  the forum’s entire decision value is gone even though  $1 - \pi$  of posts are genuine.

*Proof.* (a) is Proposition 1 with (1); (b) immediate. (c) The floor is attained:  $\mathbb{E}[M \mid p] \geq |2p - 1|/a$  always (Theorem 2), with equality everywhere once  $h \leq c_r$ , which occurs for every  $p$  when  $Q$  is close enough to  $\frac{1}{2}$ ; so as  $\pi \rightarrow \bar{\pi}(\frac{1}{2})$  the entire margin process converges to the no-forum benchmark. Pointwise monotonicity in  $Q$ : for  $c_r = 0$ ,  $a\mathbb{E}[M|p] = (4d^2 + \kappa x)/(1 + \kappa x)$  in the window and  $\partial(a\mathbb{E})/\partial\kappa = x(1 - 4d^2)/(1 + \kappa x)^2 \geq 0$  with  $\kappa = (2Q - 1)^2/[Q(1 - Q)]$  increasing in  $Q$  — proved; for  $c_r > 0$  the two channels ( $\kappa_c \uparrow$ ,  $\kappa \uparrow$ ) push in opposite directions and we verify  $\partial\mathbb{E}[M|p]/\partial Q \geq 0$  numerically (grid over  $a \in \{1, 1.5, 3\}$ ,  $c_r \leq 0.2$ ,  $p \in [0.5, 0.99]$ , 2,000 values of  $Q$ : zero violations). (d) The window condition is monotone in  $Q$  and  $c_r$ .  $\square$

Part (a) is the precise form the paper’s Proposition 2 (“unverifiability threshold”) takes in the rebuilt model, and the honest statement is slightly different from the draft’s:

- *At the individual level* the paper’s discreteness survives exactly: each reader’s problem is binary, and the marginal reader’s rule is (7) with a hard threshold.
- *At the aggregate level*, readership declines continuously and hits *exactly zero at an interior*  $\bar{\pi} < 1$  — a collapse at finite contamination, but a kinked continuous decline rather than a jump. The genuinely *discontinuous* collapse the abstract advertises is restored — and made robust — once contamination is endogenous: Theorem 4.
- The draft’s distinction between expected contamination  $\pi$  (the object in the reading rule) and realized share  $\lambda$  carries over verbatim: the decision-relevant  $\pi$  is the reader’s forward-looking expectation about the post she is about to read, which jumps at a capability release ahead of  $\lambda$ . This interpretation layer is unchanged by the rebuild and remains the mechanism for collapse at  $\lambda \approx 8\%$ .

## 9 Endogenous contamination: a tipping theorem

Close the loop: contamination depends on how much humans post, which depends on the audience, which depends on contamination. Fix a proposal type (contestedness  $x$ ; think of the representative contested proposal,  $x = \frac{1}{4}$ ) and let AI post mass be  $z \geq 0$ . An equilibrium is a triple  $(A, n, \pi)$  with

$$n = F(RA), \quad \pi = \frac{z}{z + n}, \quad A = \mathcal{A}(\pi) \equiv \frac{(2x(2q - 1)(1 - \pi) - c_r)_+}{2a D(\pi)},$$

$D(\pi) = Q(1 - Q) + x(2Q - 1)^2$  at  $Q = \frac{1}{2} + (1 - \pi)(q - \frac{1}{2})$ . Equivalently, a fixed point of the audience map  $T_z(A) \equiv \mathcal{A}(z/(z + F(RA)))$ .

**Theorem 4** (Discontinuous, hysteretic collapse [PROVED UNDER ASSUMPTIONS] (representative proposal; exogenous  $z$ ; Assumptions 1–6)). *Let  $c_r \in (0, I_H)$  and  $\bar{\pi} = 1 - c_r/I_H$ . Then:*

- (a) *Collapse always coexists.*  $(A, n, \pi) = (0, 0, 1)$  is an equilibrium for every  $z > 0$ .
- (b) *Dead zone.*  $T_z(A) = 0$  for all  $A \leq A_z \equiv F^{-1}(\min\{z^{\frac{1-\bar{\pi}}{\bar{\pi}}}, 1\})/R$ , and  $A_z > 0$  for every  $z > 0$ : audiences below  $A_z$  cannot sustain enough human posting to keep expected reading value positive.

- (c) *Tipping point.*  $T_z$  is nondecreasing in  $A$  and strictly decreasing in  $z$ , so  $z^* \equiv \sup\{z : \text{a healthy equilibrium } A > 0 \text{ exists}\}$  is well defined, with  $0 < z^* \leq \frac{\bar{\pi}}{1-\bar{\pi}} n_{\max}$  ( $n_{\max} = F(\bar{c})$ ): once AI post mass exceeds  $\frac{\bar{\pi}}{1-\bar{\pi}}$  times maximal human volume, only collapse survives. For every  $z \in (0, z^*)$  the healthy and collapsed equilibria coexist (bistability); for  $z > z^*$  collapse is unique.
- (d) *Discontinuity.* Every healthy equilibrium satisfies  $A > A_z$ ; hence as  $z$  crosses  $z^*$  the audience jumps from at least  $A_{z^*} > 0$  to 0 — never a continuous fade-out.
- (e) *Hysteresis.* Under best-response dynamics  $A_{t+1} = T_z(A_t)$ , the collapsed state is locally absorbing for every  $z > 0$  (by (b)); after a collapse, reducing  $z$  — even nearly to zero — does not restore the forum. Recovery requires coordinated re-entry of audience mass exceeding  $A_z$ , not merely undoing the shock.
- (f) *Collapse below the reading threshold.* At any healthy equilibrium, realized contamination satisfies  $\pi = z/(z+n) < \bar{\pi}$  strictly. The forum therefore always tips at a realized AI share strictly below the level  $\bar{\pi}$  at which reading value would mechanically reach zero.

*Proof.* (a)  $A = 0 \Rightarrow n = F(0) = 0 \Rightarrow \pi = 1 \Rightarrow Q = \frac{1}{2} \Rightarrow \mathcal{A} = 0$ . (b)  $\mathcal{A}(\pi) > 0$  iff  $\pi < \bar{\pi}$  iff  $F(RA) > z(1-\bar{\pi})/\bar{\pi}$ . (c) Monotonicity of  $T_z$  in  $A$ : the numerator of  $\mathcal{A}$  is increasing in  $1-\pi$  and  $\pi$  is decreasing in  $A$ ; the denominator  $D$  is decreasing in  $Q$  since  $\partial D/\partial Q = 2(2Q-1)(4x-1) \cdot \frac{1}{2} \leq 0$  for  $x \leq \frac{1}{4}$ ; strict decrease in  $z$  is immediate. Finiteness of  $z^*$ : a healthy equilibrium needs  $F(RA) > z(1-\bar{\pi})/\bar{\pi}$  with  $F \leq n_{\max}$ . Positivity: at  $z = 0$  the healthy audience is  $A_0 = \mathcal{A}(0) > 0$  (Assumption 6); for any  $A' < A_0$ ,  $T_z(A') \rightarrow \mathcal{A}(0) > A'$  as  $z \rightarrow 0$ , and a nondecreasing self-map of  $[A', A_{\max}]$  has a fixed point (Tarski); the set of  $z$  with a healthy equilibrium is a down-set by pointwise monotonicity in  $z$ . Bistability on  $(0, z^*)$ : (a) plus the down-set property. (d) By (b), any nonzero fixed point exceeds  $A_z$ , and  $A_z$  is nondecreasing in  $z$ . (e)  $T_z \equiv 0$  on  $[0, A_z]$ ,  $A_z > 0$ . (f)  $A = T_z(A) > 0$  requires  $\pi(A) < \bar{\pi}$  by (b).  $\square$

*Remark 5* (What this strengthens, and what it does not deliver). The working notes' Proposition 3 obtained tipping and hysteresis *only under an S-shaped adoption curve* and flagged that with concave adoption the decline is smooth. Theorem 4 removes that condition: a positive reading cost alone creates the dead zone that makes collapse discontinuous, locally absorbing, and coexisting with health for every  $z \in (0, z^*)$  — for *any* continuous strictly increasing  $F$ . The reading cost that generates the unverifiability threshold is the same primitive that makes the collapse irreversible; that unification is new and, we think, worth a proposition in the paper (or the lead result of a spin-off). Numerical benchmark ( $x = \frac{1}{4}$ ,  $q = 0.75$ ,  $a = 1.2$ , uniform  $F$  on  $[0, 0.25]$ ,  $R = 1$ ):  $c_r = 0.02$  gives  $\bar{\pi} = 0.92$  but collapse at  $z^* \approx 0.86$  with realized contamination  $\pi \approx 0.69$  and an audience jump  $\approx 0.10 \rightarrow 0$ ; raising  $c_r$  to 0.08 moves collapse to realized  $\pi \approx 0.41$ . Thin reading margins mean collapse strikes far below saturation. What the theorem does *not* deliver: the observed  $\lambda \approx 8\%$  is still lower than these equilibrium contamination levels under the benchmark parameters — matching the *date* of the collapse at 8% realized share still requires the expectations channel ( $\pi$  jumping ahead of  $\lambda$  at the capability release), exactly as the paper currently argues. The theorem's contribution is that collapse at low-and-partial contamination is an equilibrium property, not a fragility of one calibration. Caveats: representative-proposal reduction (aggregation across a distribution of  $p$  is not done);  $z$  exogenous; stability/hysteresis defined for best-response dynamics.

## 10 Calibration audit: what survives of $\pi^* \geq 0.082$ and 858 words

The paper’s Section 5.5 runs entirely off the reading rule, so we audit each step against the derived model.

**Step 1: the threshold identity.**  $\pi^* = 1 - c_r/I_H$ , hence “ $\pi^*$  equals the net-surplus share of a post.” *Survives exactly* (equation (7)), now with  $I_H = 2p(1 - p)(2q - 1)$  pinned by primitives and the representative delegate mapped to the peak (ex-ante indifferent) reader on the representative contested proposal. [PROVED]

**Step 2: the bound  $\pi^* \geq \lambda_{\text{break}} = 0.082$ .** The logic (reading was still happening at the break  $\Rightarrow \pi \leq \pi^*$ ; expectations run ahead of a rising realized share  $\Rightarrow \pi \geq \lambda$ ) involves no structure beyond the threshold rule and is unchanged. *Survives.* Theorem 4(f) adds an independent reason collapse precedes saturation. [PROVED UNDER ASSUMPTIONS] (the  $\pi \geq \lambda$  step is an assumption about expectations, as the paper already flags)

**Step 3: per-post surplus  $I_H - c_r \geq \pi^* I_H \geq 0.082 \times 237 \approx 19.4$  words.** Survives for the marginal reading decision of the peak reader with the paper’s normalization  $I_H = c_p = 237$  words. [PROVED UNDER ASSUMPTIONS] (normalization)

**Step 4: aggregation  $\times \bar{n} = 44.2$  posts to  $\geq 858$  words/delegate/proposal, and  $\times K = 62$  to  $\approx 43,000$ .** *This is where the derived model withholds full support.* In the one-post sampling protocol a reader consumes one post, and the derived aggregate surplus is far smaller. If delegates read whole threads (the paper’s implicit protocol), per-post surplus is constant at 19.4 only if successive posts do not become redundant. With conditionally i.i.d. signals, marginal reading value *declines* in posts already read, and the sum is strictly less than  $44.2 \times 19.4$ ; only the first-post bound ( $\geq 19.4$  words per delegate per proposal) is fully derived. The 858 figure survives under:

**Assumption 7** (Non-redundancy, NR). *Posts address (approximately) distinct payoff-relevant aspects of a proposal, so the informational value of each successive post is undiminished and the threshold calculus (7) applies post-by-post.*

Under NR the paper’s arithmetic goes through unchanged. NR is defensible for governance threads (posts flag distinct risks, precedents, technical issues) but it is an assumption, not a theorem; a redundant-signal version of the same calculation would justify only the 19.4-word floor plus unquantified inframarginal surplus. Additionally, averaging over the reading band rather than the peak reader halves per-reader surplus (tent geometry). *Verdict: 858 and 43,000 are [PROVED UNDER ASSUMPTIONS] (NR + whole-thread protocol + peak-reader mapping) — i.e., exactly the “illustrative calibration under stated assumptions” framing of agenda Decision 2, which this analysis independently recommends. The units, the bound logic, and the 19.4 floor are solid; the headline magnitudes are assumption-carrying.*

*Remark 6* (Multi-post reading [SKETCH]). A full sequential-reading extension (read posts one at a time at  $c_r$  each, stop optimally) would endogenize how many of the  $\bar{n}$  posts are read and let the data separate NR from redundancy via the observed reads-per-post decay within threads. This is the natural next modeling increment if the calibration is to be upgraded from illustrative to identified; it is not done here.

## 11 Model–data audit

Where the empirics constrain the theory, does the derived model deliver?

| Empirical shape                                                   | Delivered?                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Posting volume peaked on contested proposals ( $\propto p(1-p)$ ) | Yes: exact closed form (6)–(8), $\propto p(1-p)$ to first order. Locus not separable from linear reduced form at $n = 29$ — claim the sign, not the shape.                                                                                     |
| Margin increasing in consensus ( $\propto  2p-1 $ )               | Yes: Theorem 2; exact floor $ 2p-1 /a$ , strict monotonicity everywhere.                                                                                                                                                                       |
| Cov(posts, margin) $< 0$ pre-AI; $\rightarrow 0$ post             | Yes: Theorem 3; post-collapse both $n$ and the $p$ -sensitivity of $M$ shut down, so the covariance dies.                                                                                                                                      |
| Collapse at low realized AI share                                 | Qualitatively yes: interior threshold $\bar{\pi} < 1$ ; equilibrium tipping strictly below $\bar{\pi}$ (Theorem 4f); the specific 8% still needs the expectations channel $\pi > \lambda$ .                                                    |
| Reads per post decline post-shock                                 | Yes, new derived prediction: $A$ near-linear in $(1-\pi)$ . Empirically right sign, underpowered (working notes §6.1).                                                                                                                         |
| Delegates using AI hold less voting power                         | Outside the formal model. Consistent with the attention channel (uninformative posts attract fewer readers/delegators) — narrative only. The career-concerns reading is <i>ruled out</i> by Theorem 1.                                         |
| <b>Honest misses (must be acknowledged in the paper):</b>         |                                                                                                                                                                                                                                                |
| Post-break margins <i>rose</i> (0.72 $\rightarrow$ 0.91)          | Model predicts compression toward $ 2p-1 /a$ on contested proposals, not a rise. A rise requires a composition shift in the $p_i$ distribution (fewer contested proposals reaching votes post-shock) — plausible, untested, outside the model. |
| Humans still write $\approx 8.9k$ words/proposal post-break       | Model's $n \rightarrow 0$ at collapse overshoots. Patch: an audience-independent posting motive $\varepsilon > 0$ (intrinsic/expressive) leaves all propositions intact and delivers residual volume; not formalized here.                     |

## 12 Verdict for the Thursday meeting

### Fully proved (from primitives, given the model's stated assumptions)

- Interior margins with closed forms; expected margin strictly increasing in consensus, floor  $|2p-1|/a$  (Prop. 3, Thm. 2).
- Readership tent and closed form; the paper's reading rule  $V(\pi) = (1-\pi)I_H - c_r$  and threshold  $\pi^* = 1 - c_r/I_H$ , with  $I_H = 2p(1-p)(2q-1)$  (Prop. 1).
- The paper's Proposition 1, Cov(posts, margin)  $< 0$  (Thm. 3).
- Career concerns as posting motive are wrong-signed for any increasing reputation payoff — the roadmap's step (ii) is refuted and replaced (Thm. 1).
- Exact welfare: forum value =  $\min(p, 1-p)$  on an interior window (Prop. 4). The old kernel  $4p(1-p)(q-\frac{1}{2})^2$  is *not derivable* and must be replaced or demoted to a heuristic.

- AI comparative statics: dilution equation from the sampling protocol; window-first welfare collapse; margin compression; readership near-linear in  $(1 - \pi)$  (Prop. 5).

### Proved under stated assumptions

- The whole edifice rests on Assumptions 2 (decision-theoretic voting/reading) and 4 (attention rewards): these *bypass* pivotality rather than solve it. They are standard, defensible, and must be stated in the paper as assumptions with their microfoundations discussed — this is the honest answer to the referee, and it is a good one: given two behavioral primitives, everything else is derived.
- Truthful posting (Assumption 3; reputational cheap-talk justification, not modeled).
- Tipping Theorem 4: discontinuous, hysteretic, below-threshold collapse — proved, under a representative-proposal reduction with exogenous AI mass and best-response dynamics. Strictly stronger than the S-shape-conditional sketch in the working notes.
- Calibration: threshold identity and  $\pi^* \geq 0.082$  survive; the 19.4 words/post surplus floor survives; **858 words/delegate and 43,000 words/proposal survive only under Assumption NR** (non-redundant posts + whole-thread reading). Present as “illustrative under stated assumptions” (confirms agenda Decision 2’s recommendation).

### Open / sketch

- Multi-post sequential reading (would upgrade the calibration; Remark 6).
- Aggregating the tipping theorem across a cross-section of proposals; endogenous AI supply  $z$ ; equilibrium inference from silence in the career-concerns game.
- The post-break *rise* in margins and residual human posting: need a composition story and an  $\varepsilon$ -posting motive respectively; neither is formalized.

### Recommendation

“Derive fully” is **viable for the core** and no longer risky: the derivations above are complete, checked numerically end-to-end, and produce the paper’s Propositions 1 and 2 as theorems. The honest package for the paper is a hybrid:

1. *Replace §5.1–5.3 with the derived two-tier model* (Sections 2–6, 8 here), stating Assumptions 2 and 4 candidly. Est. 1–1.5 weeks of writing; no new mathematical risk remains.
2. *Replace the welfare formula* with Proposition 4 (or demote the monomial to an explicit heuristic). Mandatory — the old formula is now known (to us) to be underivable, and a referee can find this.
3. *Reframe the calibration* as illustrative under stated assumptions, keeping the threshold identity and the 0.082 bound as the load-bearing numbers and stating NR when quoting 858/43,000. (= agenda Decision 2.)
4. *Tipping theorem*: include compactly as the mechanism capstone (it is now a theorem, not a possibility sketch) *or* hold it for the theory follow-up; either is defensible. If included, keep the expectations-vs-realization distinction as the bridge to the 8% fact.

5. *Do not* use career-concerns language for the posting motive or the delegate cross-section anywhere in the paper; Theorem 1 makes that framing indefensible.

Estimated total: 2–3 weeks to a referee-solid Section 5, most of it exposition rather than mathematics. The alternative (“reframe as illustrative”) is no longer the safer option: the derivation exists, is checkable, and the illustrative framing would leave the strongest referee line unanswered while this note shows it answerable.

## References

- [1] Akerlof, G. (1970). The market for “lemons”: Quality uncertainty and the market mechanism. *Quarterly Journal of Economics*, 84(3), 488–500.
- [2] Dewatripont, M., and J. Tirole (1999). Advocates. *Journal of Political Economy*, 107(1), 1–39.
- [3] Feddersen, T., and W. Pesendorfer (1997). Voting behavior and information aggregation in elections with private information. *Econometrica*, 65(5), 1029–1058.
- [4] Feddersen, T., and A. Sandroni (2006). A theory of participation in elections. *American Economic Review*, 96(4), 1271–1282.
- [5] Grossman, S., and J. Stiglitz (1980). On the impossibility of informationally efficient markets. *American Economic Review*, 70(3), 393–408.
- [6] Holmström, B. (1999). Managerial incentive problems: A dynamic perspective. *Review of Economic Studies*, 66(1), 169–182.
- [7] Judd, K. (1985). The law of large numbers with a continuum of IID random variables. *Journal of Economic Theory*, 35(1), 19–25.
- [8] Lohmann, S. (1994). Information aggregation through costly political action. *American Economic Review*, 84(3), 518–530.
- [9] Ottaviani, M., and P. Sørensen (2006). Reputational cheap talk. *RAND Journal of Economics*, 37(1), 155–175.